COMPUTER NETWORKS – ASSIGNMENT 1

CO1 | Task 1 – Size the Traffic Before Choosing Anything

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Assignment: Make the Data of a Smart Manufacturing Plant Arrive Correctly and On Time

1. Calculate the traffic of each source

Traffic source	Calculation	Traffic
Robot controllers	40 × (200 × 8 bits ÷ 0.002 s)	32 Mbps
Vibration sensors	600 × 50 bytes/s × 8 bits	0.24 Mbps
CCTV cameras	24 × 4 Mbps	96 Mbps
Office users	150 × 2 Mbps × 10% active	30 Mbps

Total traffic per production line: 32 + 0.24 + 96 + 30 = **158.24 Mbps**.

Total traffic for 8 lines: 158.24 × 8 = **1,265.92 Mbps** ≈ **1.266 Gbps**.

2. Highest data source and strictest delay requirement

CCTV sends the most data at 96 Mbps per line. Robot controllers have the strictest delay requirement because their commands must arrive within 10 milliseconds. Therefore, the source sending the most data and the source with the strictest delay requirement are not the same.

3. Is a 1 Gbps uplink enough?

For one line, 158.24 Mbps is below 1,000 Mbps, so a 1 Gbps uplink is sufficient for the calculated traffic. For the complete plant with 8 lines, 1,265.92 Mbps is greater than 1,000 Mbps. Therefore, one 1 Gbps uplink is not sufficient for all 8 lines at peak traffic. The plant should use a higher-capacity backbone or multiple uplinks, with extra capacity for protocol overhead and future growth.

If CCTV and robot commands share the same queue, heavy video traffic can create congestion and queueing delay. Robot commands may miss their 10 ms deadline. Use separate VLANs and quality-of-service (QoS) queues, giving robot-control traffic the highest priority.

4. Ranking by harm from a 100 ms delay

Rank	Traffic type	Reason
1	Robot commands	A 100 ms delay is ten times the 10 ms deadline and may affect safe, timely machine control.
2	Vibration sensors	Delayed readings can slow detection of machine vibration or faults.
3	CCTV video	A delay can make live monitoring less immediate, although recording can be buffered.
4	Office traffic	A short delay usually affects email or ERP responsiveness but is less critical to machine control.

This ranking assumes that the sensor readings are used for plant monitoring and that CCTV is recorded locally or can tolerate a small playback delay. The robot deadline is explicitly given in the question.

5. Alternative concept: CCTV sends motion-triggered clips only

CCTV traffic becomes 5% of its continuous traffic: 96 × 0.05 = **4.8 Mbps** per line.

New traffic per line = 32 + 0.24 + 4.8 + 30 = **67.04 Mbps**.

New traffic for 8 lines = 67.04 × 8 = **536.32 Mbps**.

Since 536.32 Mbps is below 1 Gbps, the 1 Gbps uplink is sufficient for the calculated eight-line traffic in this alternative. It is no longer the traffic bottleneck under these assumptions, but capacity should still be reserved for overhead, bursts and future growth.

Conclusion

CCTV produces the highest traffic, but robot commands need the lowest delay. A single 1 Gbps uplink cannot carry the calculated peak traffic of all eight lines when CCTV streams continuously. Recording CCTV locally and sending only motion-triggered clips reduces the eight-line traffic to 536.32 Mbps. QoS should protect robot commands from delay caused by other traffic.